Exact Renewal Structure in Accelerated Collatz Dynamics

A 2-adic Symbolic Model and Exact Operator Theory

Hiroki Kamanoi

2026

Abstract

We construct an exact symbolic dynamical model associated with the accelerated Collatz first-return map on the Type A congruence class

$$\mathcal{A} := \{n \in \mathbb{Z}_{>0} : n \equiv 3 \pmod 4\}.$$

The paper separates the problem into two logically distinct layers.

First, we define an autonomous countable-state renewal system

$$(X, \sigma, \pi)$$

and establish its exact operator structure. The associated Markov operator admits an exact shift-and-kill decomposition on mean-zero observables:

$$P^N f = S^N f.$$

From this identity we derive exact formulas for covariance decay, spectral contraction, entropy, and total mutual information memory.

Second, we show that accelerated Collatz dynamics embeds densely into this symbolic system through a 2-adic coding map.

The present work does not prove the Collatz conjecture. Its purpose is to isolate and rigorously characterize the symbolic and operator-theoretic structure underlying the induced Type A dynamics.

Contents

1 Introduction

The accelerated Collatz map is defined by

$$T(n) = \begin{cases} n/2 & (n \text{ even}), \\ (3n+1)/2^{v_2(3n+1)} & (n \text{ odd}). \end{cases}$$

Rather than studying arbitrary orbits directly, we focus on the induced first-return dynamics on the congruence class

$$\mathcal{A} = \{n \equiv 3 \pmod 4\}.$$

The key observation is that the valuation coordinate

$$k(n) := v_2(n+1) - 1$$

evolves according to a renewal mechanism:

- deterministic descent for $k \geq 2$,

- geometric renewal at $k = 1$.

The main contribution of this paper is the separation between:

(i) an autonomous symbolic renewal system,

(ii) an arithmetic realization arising from Collatz dynamics.

The symbolic system itself admits a completely explicit operator theory.

1.1 Notation and Dynamical Conventions

We distinguish carefully between:

(1) arithmetic Collatz dynamics,

(2) the autonomous renewal system,

(3) the 2-adic symbolic realization.

This separation is maintained throughout the paper.

Symbol	Meaning
T	accelerated Collatz map
$T_{\mathcal{A}}$	first-return map on Type A integers
X	renewal shift
σ	left shift
P	Markov operator
S	shift-and-kill operator
π	stationary distribution
$k(n)$	valuation coordinate $v_2(n+1) - 1$

1.2 Topological Structure

The renewal shift X is endowed with the product topology induced from the discrete topology on $\mathbb{N}$.

The completion

$$\widehat{\mathcal{A}} = \{x \in \mathbb{Z}_2 : x \equiv 3 \pmod 4\}$$

inherits the standard 2-adic topology.

Finite symbolic words correspond to finite congruence conditions modulo powers of 2.

2 Arithmetic Setup

Define

$$\mathcal{A} = \{n \in \mathbb{Z}_{>0} : n \equiv 3 \pmod 4\}.$$

Let

$$T_{\mathcal{A}}$$

denote the first-return map to $\mathcal{A}$.

For $n \in \mathcal{A}$, define

$$k(n) = v_2(n+1) - 1.$$

3 The Autonomous Renewal System

3.1 State Space

Define the renewal shift

$$X \subset \mathbb{N}^{\mathbb{N}}$$

generated by transitions

$$j \to j-1 \qquad (j \geq 2),$$

and

$$1 \to m \qquad (m \geq 1).$$

3.2 Markov Kernel

Define the transition kernel

$$P(j, j-1) = 1 \qquad (j \geq 2),$$

and

$$P(1, m) = 2^{-m}.$$

Theorem 3.1 (Stationary Measure)**.** *The probability distribution*

$$\pi(m) = 2^{-m}$$

is stationary and unique.

Proof. For $m \geq 1$,

$$(\pi P)(m) = \pi(m+1) + \pi(1)2^{-m} = 2^{-m}.$$

Irreducibility and positive recurrence imply uniqueness. □

4 Operator Structure

4.1 Markov Operator

Define

$$(Pf)(j) = \sum_m P(j, m) f(m).$$

Explicitly,

$$(Pf)(j) = f(j-1) \qquad (j \geq 2),$$

and

$$(Pf)(1) = E_\pi[f].$$

4.2 Shift-and-Kill Operator

Define

$$(Sf)(j) = f(j-1)\mathbf{1}_{j \geq 2}.$$

Theorem 4.1 (Shift-and-Kill Identity)**.** *For every mean-zero function $f \in L^2(\pi)$,*

$$P^N f = S^N f.$$

Proof. For $N = 1$,

$$(Pf)(j) = f(j-1) = (Sf)(j) \qquad (j \geq 2),$$

while

$$(Pf)(1) = E_\pi[f] = 0 = (Sf)(1).$$

Since P preserves the mean-zero subspace, induction yields

$$P^{N+1} f = S^{N+1} f.$$

□

5 Exact Mixing and Spectral Theory

Theorem 5.1 (Exact Covariance)**.** *Let*

$$g(k) = k - 2.$$

Then

$$\mathrm{Cov}_\pi(k_0, k_N) = 2^{1-N}.$$

Proof. By the shift-and-kill identity,

$$\mathrm{Cov}_\pi(k_0, k_N) = \langle g, S^N g \rangle_\pi.$$

Direct summation gives

$$\sum_{j>N} (j-2)(j-N-2)2^{-j} = 2^{1-N}.$$

□

Theorem 5.2 (L^∞-Mixing)**.** *For bounded observables f, g,*

$$|\,\mathrm{Cov}_\pi(f(k_0), g(k_N))| \leq \|f\|_\infty \|g\|_\infty 2^{-N}.$$

Theorem 5.3 (L^2 Spectral Decay)**.** *For mean-zero $f \in L^2(\pi)$,*

$$\|P^N f\|_{L^2(\pi)} = 2^{-N/2} \|f\|_{L^2(\pi)}.$$

6 Entropy and Information

Theorem 6.1 (Entropy)**.** *The measure-theoretic entropy satisfies*

$$h(\pi, \sigma) = \log 2.$$

Theorem 6.2 (Total Mutual Information)**.**

$$\sum_{N=1}^{\infty} I_\pi(k_0; k_N) = 2 \log 2.$$

7 Arithmetic Realization

Define

$$\widehat{\mathcal{A}} = \{x \in \mathbb{Z}_2 : x \equiv 3 \pmod 4\}.$$

Theorem 7.1 (2-adic Symbolic Realization)**.** *The coding map*

$$\widehat{\Phi} : \widehat{\mathcal{A}} \to X$$

is a homeomorphism.

Positive integers inject densely into X through this realization.

Proof. We divide the proof into four parts.

Injectivity. Finite symbolic prefixes determine congruence classes modulo powers of 2. Hence two points with identical symbolic itineraries agree modulo 2^N for every N, and therefore coincide in $\mathbb{Z}_2$.

Surjectivity. Given an admissible symbolic sequence

$$(k_1, k_2, \ldots) \in X,$$

the finite prefix $(k_1, \ldots, k_r)$ determines a nonempty congruence class modulo 2^{M_r}, where

$$M_r = k_1 + \cdots + k_r + r.$$

These congruence classes define nested 2-adic balls

$$B_1 \supset B_2 \supset \cdots$$

with diameters tending to zero.

Compactness and completeness of $\mathbb{Z}_2$ imply

$$\bigcap_r B_r$$

contains a unique point.

Continuity. Cylinder sets in X correspond exactly to finite congruence conditions modulo powers of 2. Hence inverse images of cylinders are open in the 2-adic topology.

Continuity of the inverse. Finite 2-adic congruence information determines finite symbolic prefixes, so the inverse map is continuous. □

8 Scope and Limitations

This paper establishes:

- an exact autonomous renewal system,
- its operator structure,
- exact covariance and entropy formulas,
- a 2-adic arithmetic realization.

The paper does not prove the Collatz conjecture.

In particular, no statement is made concerning:

- global convergence,
- divergent orbit exclusion,
- orbitwise ergodicity,
- arithmetic decorrelation.

9 Conclusion

The induced Type A dynamics admits an exact symbolic renewal structure whose operator theory is completely computable.

The remaining obstruction appears not operator-theoretic, but arithmetic: whether an individual orbit can sustain persistent deviations from the renewal background.

A Numerical Sanity Checks

Finite computations confirm:

- exact covariance values for small N,
- symbolic congruence reconstruction,
- renewal transition statistics.

These computations are included only as consistency checks and are not used in any proof.

B Entropy and KL Computations

The mutual information formula follows from the deterministic survival of non-renewed trajectories:

$$I_\pi(k_0; k_N) = 2\log 2 \cdot 2^{-N}.$$

Summing the geometric series yields

$$\sum_{N\geq 1} I_\pi(k_0; k_N) = 2\log 2.$$

References

[1] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi (2019).

[2] P. Walters, *An Introduction to Ergodic Theory*, Springer, 1982.

[3] B. Kitchens, *Symbolic Dynamics*, Springer, 1998.

[4] D. Lind and B. Marcus, *An Introduction to Symbolic Dynamics and Coding*, Cambridge University Press.

[5] O. Sarig, *Thermodynamic formalism for countable Markov shifts*, Ergodic Theory and Dynamical Systems.

[6] J. Aaronson, *An Introduction to Infinite Ergodic Theory*, AMS.